

SAKET TRIPATHI

AI/ML Engineer | Deep Learning | NLP | Computer Vision | Research

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PROFESSIONAL SUMMARY

ML Engineer with hands-on research experience at IIT Roorkee and MANIT Bhopal, with research presented at ICICDS-2026 (Thailand), soon to be published. Demonstrated ability to design and ship end-to-end ML pipelines achieving 97.01% accuracy on handwriting recognition (CNN-BiLSTM) and 2.27% MAPE on Indian stock prediction (Fusion-LSTM + FinBERT), outperforming all benchmarks. Proficient in PyTorch, TensorFlow, HuggingFace Transformers, LangChain, and full-stack ML deployment. Seeking ML/AI/NLP internship or research roles.

EDUCATION

Integrated B.Tech + M.Tech — Computer Science & Engineering

2022 – 2027

Central University of Jharkhand, Ranchi | CGPA: 8.55 / 10 (up to 8th Semester)

Class XII — ICSE Board *Completed*

St. Francis School, Varanasi | 80%

Class X — ICSE Board *Completed*

St. Francis School, Varanasi | 83%

RESEARCH & INTERNSHIP EXPERIENCE

Research Intern — AI/ML — MANIT Bhopal

Jun 2025 – Jul 2025

- Researched Indian stock price movement prediction using deep learning and FinBERT-based financial news sentiment analysis.
- Designed Fusion-LSTM model combining OHLCV time-series data with real-time news sentiment for 10 Indian stocks (2022–2025).
- Achieved MAE 33.41 vs. 45.73 baseline and MAPE 2.27% vs. 3.44% — outperformed all benchmark models including vanilla LSTM and Autoencoder.
- Research presented at ICICDS-2026 (Thailand); soon to be published.

Machine Learning Intern — Computer Vision & NLP — IIT Roorkee

Sep 2024 – Feb 2025

- Developed a cursive handwriting recognition system using CNN-BiLSTM architecture trained on 80,850 labeled image samples.
- Built a scalable preprocessing pipeline with image normalization, augmentation, and sequence modeling for OCR tasks.
- Achieved 97.01% test accuracy (loss: 0.1214), significantly outperforming traditional OCR baselines.
- Collaborated with senior researchers on model optimization and hyperparameter tuning using TensorFlow/Keras.

KEY PROJECTS

Stock Price Movement Prediction — Fusion-LSTM + FinBERT | Python, LSTM, FinBERT, TensorFlow, NLP, Pandas

- Fused OHLCV stock data with FinBERT sentiment features for 10 Indian stocks (2022–2025) in a multi-source LSTM model.
- Outperformed Autoencoder baseline: MAE 33.41 vs. 45.73, MAPE 2.27% vs. 3.44% — research presented at ICICDS-2026, soon to be published.
- End-to-end ML pipeline: data collection, preprocessing, feature engineering, training, evaluation, and benchmarking.

Repository: github.com/Tripathijii147/Stock-Price-Prediction-model

Psychological Stress Detection from Space Audio — Ensemble ML | Python, librosa, XGBoost, LightGBM, Stacking Ensemble, Scikit-learn

- Built a stress-detection pipeline trained on RAVDESS (2,880 recordings) and evaluated on real NASA Apollo mission audio — novel cross-dataset evaluation.
- Engineered 262-dimensional acoustic feature set; applied consensus feature selection reducing to 60 dims, improving F1 from 0.7818 to 0.7897.
- Stacking Ensemble achieved 83.8% accuracy, F1=0.8365, AUC-ROC=0.9146; up to 67.9% on Apollo audio with zero Apollo training data.

Repository: github.com/Tripathijii147/stress-detection-space-audio

Autism Spectrum Disorder Detection — Multimodal Fusion | Python, BERT, CNN, librosa, TensorFlow, SVM, TF-IDF

- Benchmarked four architectures across text and audio modalities using late and early fusion strategies.
- Best result: TF-IDF + Audio + SVM achieved 85% accuracy and F1=0.86; BERT fine-tuned text model achieved ROC-AUC=0.9423, PR-AUC=0.9499.
- End-to-end multimodal pipeline evaluated using MCC, Cohen Kappa, and Brier Score for rigorous comparison.

Repository: github.com/Tripathiji147/autism-detection-multimodal

Drug-Target Interaction (DTI) Prediction | *Python, Scikit-learn, PyTorch, RDKit, Pandas*

- Built ML/DL classification models to predict drug-protein interactions using molecular descriptors and protein sequence features.
- Evaluated using AUC-ROC, Precision, Recall, and F1-score — demonstrates cross-domain AI application for biotech/healthtech.

PUBLICATION

Chaurasiya R.K., Singh S., Tripathi S. — Stock Movement Prediction using Auto-Encoder and Fusion-LSTM with News Sentiments. Presented at the Third International Conference on Innovations in Cybersecurity and Data Science (ICICDS-2026), S.E.A. College of Engineering & Technology and Shinawatra University, Thailand, 25–27 June 2026. Soon to be published.

TECHNICAL SKILLS

Languages: Python, Java, C++, SQL, MySQL

ML / DL Frameworks: PyTorch, TensorFlow, Keras, Scikit-learn, HuggingFace Transformers, FinBERT

NLP: BERT, Sentiment Analysis, Text Classification, Sequence Modeling, LangChain, RAG, Prompt Engineering

Computer Vision: CNN, CNN-BiLSTM, OpenCV, Image Augmentation, OCR

Data & Analysis: Pandas, NumPy, Matplotlib, Jupyter Notebook

Deployment / Cloud: HuggingFace Spaces, Google Colab, Streamlit, FastAPI

Web Technologies: HTML, CSS, JavaScript

Soft Skills: Research Communication, Leadership, Teamwork, Project Management

CERTIFICATIONS

- Programming in Java — NPTEL / IIT System (Jan–Apr 2024): 12-week course covering OOP, core Java, and data structures.
- SQL (Basic to Advanced) — Geekster (Jun 2024): Hands-on SQL queries, joins, subqueries, and database design.
- Ethical Hacking and Penetration Testing — Virtual Internship, C-DAC Noida (Sep–Oct 2024), under Cyber Gyan Project.
- AICTE Oasis Infobyte Internship — recognized as Star Performer (2024).
- Python Development Internship — TechOctanet Services Pvt. Ltd. (Aug–Sep 2024).

EXTRA-CURRICULAR ACTIVITIES

- Treasurer, Programming Club — Code Crafter (2022–Present): Managed finances, organized coding competitions, and promoted a coding culture on campus.
- NCC Cadet — Completed 'B' and 'C' Certificate Examinations: Trained through structured drills, camps, and national-level programs; developed leadership, discipline, teamwork, and time management skills applied directly to long-term project and research work.